USER GUIDE

DataCo Supply Chain Analytics Application

A Simple Guide to Understanding Every Tab, Analysis, Prediction and Business Decision

> An easy-to-understand guide to using and interpreting the DataCo Supply Chain Analytics decision-support application. Written for readers with no technical, statistical or machine-learning background.

Contents

1. How the application works

This application helps a supply chain company answer one practical question: which orders are likely to arrive late, and what should we do about it?

It is not just a prediction tool. Predicting that an order will be late is only useful if you can then do something about it, and if you know whether doing that something is worth the money. So the application takes you through a complete chain of reasoning: understand the data, find the patterns, check whether those patterns are real, build a prediction, work out which factors matter, identify what management can actually change, and finally price that decision.

Here is the whole journey. Each step has a tab in the application.

Business problem — What are we trying to solve? Deliveries arrive late and we want to know why, and what to do.

▼

Data — What information do we actually have? Three years of orders, with details of what was bought, where it went and how it was shipped.

▼

Data cleaning — Remove empty columns, duplicate columns and personal data, and make sure we are counting orders rather than counting the same order several times.

▼

Exploratory analysis — Look at the patterns. Which shipping methods run late? Which countries? Which products?

▼

Statistical analysis — Check whether those patterns are real differences or just random variation.

▼

Prediction model — Train a model to estimate the chance that a given order will arrive late.

▼

Identify key risk drivers — Work out which pieces of information the model is actually relying on.

▼

Shipping mode decision — Of everything that predicts lateness, which can management actually change? The shipping method.

▼

Economic impact — Faster shipping costs more. Late delivery costs goodwill. Which trade-off is better?

▼

Scenario and what-if analysis — Our cost estimates might be wrong. Does the recommendation survive if they are?

▼

Management recommendation — A specific, defensible course of action, with its assumptions stated.

IN PLAIN ENGLISH

We first understand the data, then look for patterns in delivery delays, then build a model that predicts delay risk. Once we know which orders are risky, we use that information to work out which shipping option makes the most economic sense — and we check whether that answer changes if our cost assumptions turn out to be wrong.

What the application is built on

Everything in the application comes from the DataCo supply chain dataset: three years of real order records from 1 January 2015 to 31 January 2018. Nothing on any screen is illustrative or made up. Every percentage, chart and model score is calculated from that data when the application is prepared, and the same numbers appear everywhere they are referenced.

The one exception is clearly and repeatedly flagged. The dataset contains no shipping-cost information at all — no freight charges, no carrier invoices. So when the application needs to compare the cost of two shipping options, it uses figures that you supply and labels them as assumptions rather than facts. This is explained fully in Tab 9.

2. The business problem in plain English

The formal question the application addresses is:

> **THE BUSINESS QUESTION**
>
> Which operational and external factors significantly predict delivery delays, and what is the optimal logistics network design decision under uncertainty to minimize total supply chain cost?

That sentence contains three separate ideas, and the application is built around connecting them. Understanding this connection is the single most important thing for following the rest of the guide.

The three pieces

PIECE	IN THE DATA	WHAT IT MEANS
Prediction What can we forecast?	`Late_delivery_risk`	For every order, did it arrive later than promised? This is the thing we are trying to predict before it happens. It is a simple yes/no.
Decision What can we change?	`Shipping Mode`	The shipping method chosen for the order: Standard Class, Second Class, First Class or Same Day. This is the lever management actually controls. We cannot move a customer to a different country, but we can choose how to ship to them.
Economic outcome What does it cost or earn?	`Order Profit Per Order`	The profit made on each order. A shipping decision that avoids delays but destroys margin is not a good decision, so the money has to be part of the picture.

> **THE KEY IDEA**
>
> The application is **not** only trying to predict whether an order will be late. It is trying to use that prediction to help management make a better shipping decision, while keeping an eye on the financial consequences.
>
> Think of it as three questions in sequence: Is this order likely to be late? → Would shipping it differently help? → Is that change worth what it costs?

Why "under uncertainty" matters

The business question asks for the best decision under uncertainty. That phrase is doing real work. We do not know exactly what a late delivery costs the business in lost goodwill. We do not know exactly what each shipping method costs. Different reasonable people would put different numbers on both.

So rather than producing one answer and pretending to be certain, the application lets you change those numbers and shows you how far you would have to move them before the recommendation changes. A recommendation that survives a wide range of assumptions is far more useful than a precise-looking number built on one guess.

A note on what this analysis can and cannot claim

Throughout the application you will see careful phrases like "associated with", "model-implied" and "historically observed", rather than "causes". This is deliberate.

The data records what happened, but nobody ran a controlled experiment. Orders were not randomly assigned to shipping methods. So when we see that one shipping method has more late deliveries than another, we can say the two things go together in the historical record. We cannot prove from this data alone that switching an order's shipping method would cause its delivery outcome to change.

> **WATCH OUT FOR**
>
> This distinction matters in a presentation. Saying "shipping mode is strongly associated with late delivery, and here is what the model expects if we changed it" is defensible. Saying "shipping mode causes late delivery" is a claim this analysis does not support, and a sharp examiner will notice the difference.

3. Finding your way around

The application opens in a web browser at `http://127.0.0.1:5000`. Across the top is a dark blue bar with the application name, and beneath it a row of twelve tabs.

The twelve main tabs

They are arranged in the order you would naturally work through the problem, from left to right:

TAB	WHAT IT IS FOR
Home	The headline numbers and the overall story
Data	What information we have and how clean it is
EDA	Exploring patterns in the data with charts
Statistical Tests	Checking which patterns are real
Classification	Building and comparing prediction models
Model Diagnostics	Checking how good the chosen model is
Risk Drivers	Which factors the model relies on
Shipping Mode	Comparing the four shipping options
Decision Optimization	Pricing the shipping decision
Simulator	Testing a single order interactively
Business Insights	The executive summary and recommendations
Presentation Mode	A guided walkthrough for presenting

Four more pages in the top corner

In the top-right of the dark blue bar are four smaller links: **Pipeline**, **Governance**, **Downloads** and **Upload**. These are supporting pages rather than analysis pages, and they are covered in sections 13 to 16.

If you only have five minutes

Open **Presentation Mode**. It walks through the entire project in twelve numbered steps, each explaining what is being analysed, why that method was used, what it found, what it means for the business, and what to do about it. Section 6 of this guide gives you a spoken script to go with it.

A WORD ON THE NUMBERS IN THIS GUIDE

Every figure quoted in the following sections is the actual value the application produces. If you re-run the analysis without changing anything, you will see exactly these numbers, because the process uses a fixed random seed. If you change the cost assumptions on the Decision Optimization tab, the cost figures will change — that is the point of them.

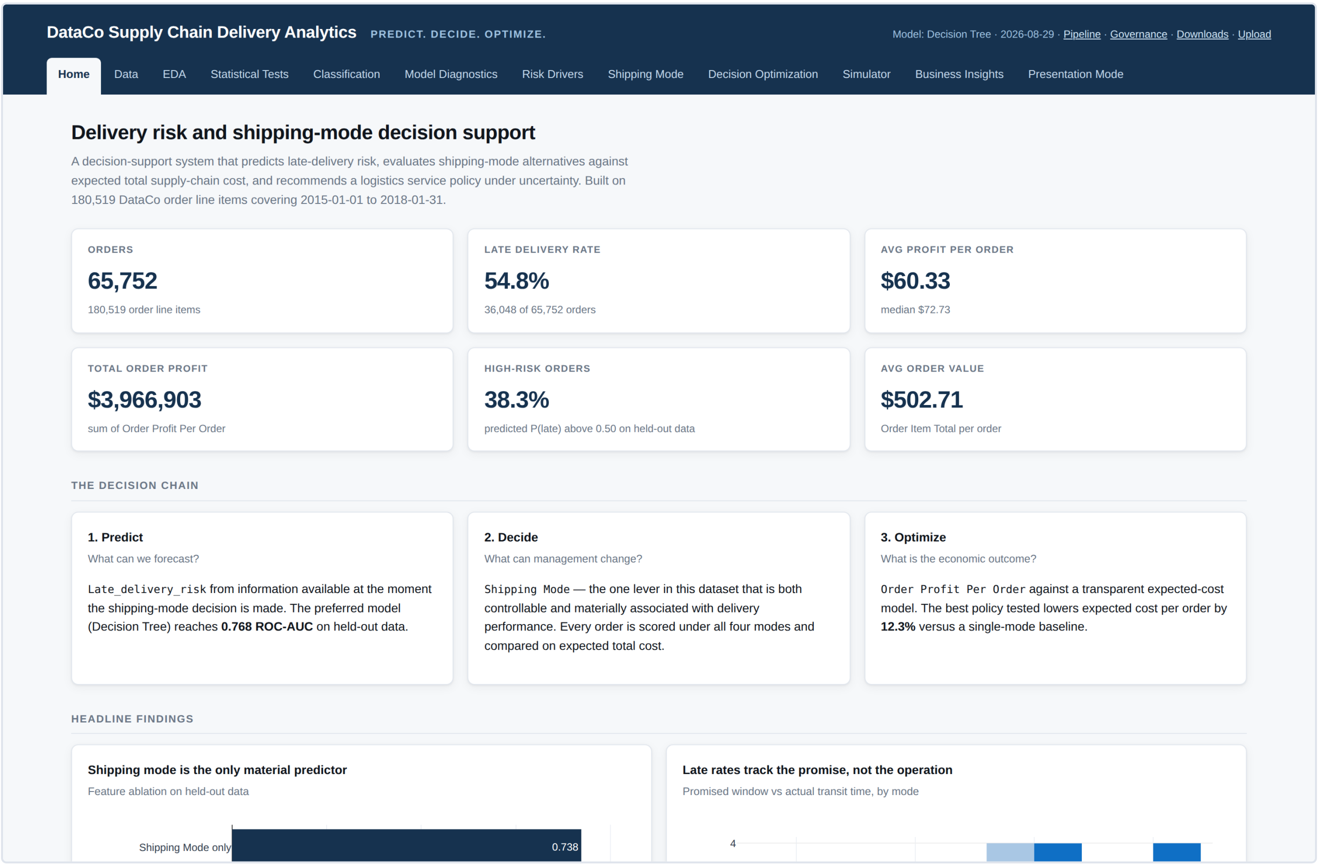
Home

A. WHAT IS THIS TAB?

This is the front page of the application. It gives you the six headline numbers about the business, explains in three short panels what the application is trying to do, and shows the two most important findings straight away. If you only look at one screen, look at this one.

B. WHY DO WE NEED THIS TAB?

It answers the very first question anyone asks: how big is this problem, and what is this tool for? A manager who has never seen the application should be able to read this page and know both within a minute.



The Home tab. Six headline figures at the top, the prediction–decision–outcome chain in the middle, and the two headline findings below.

C. WHAT DO WE SEE HERE?

Six summary cards across the top:

CARD	VALUE	WHAT IT MEANS
Orders	65,752	How many separate customer orders the analysis covers, drawn from 180,519 individual order lines. One order can contain several products.
Late delivery rate	54.8%	The share of orders that arrived later than the date promised to the customer. More than half — which is why this analysis exists.
Avg profit per order	$60.33	Average profit made on an order. The median is $72.73; the average is lower because some orders lose money and pull it down.
Total order profit	$3,966,903	All the profit across the three years, added up.
High-risk orders	38.3%	The share of orders the model flags as more likely than not to be late, when tested on data it had never seen.
Avg order value	$502.71	Average total value of an order.

Three panels — "Predict", "Decide", "Optimize" — set out the logic of the whole application: what we forecast, what management can change, and how we judge whether the change is worthwhile.

Two headline finding charts at the bottom. The left one compares how well the model predicts with and without shipping mode. The right one compares, for each shipping method, the number of days promised against the number of days actually taken.

D. HOW SHOULD I READ IT?

The left-hand chart shows three bars, each a score between 0.5 and 1.0. Think of this score as "how good is the model at telling risky orders apart from safe ones", where **0.5 is no better than tossing a coin** and 1.0 would be perfect.

- **Shipping Mode only — 0.738.** Using nothing but the shipping method.
- **All features — 0.768.** Using everything available.
- **All except Shipping Mode — 0.535.** Using everything except the shipping method: barely better than a coin toss.

The right-hand chart has two bars for each shipping method: the pale bar is the number of days promised, the dark bar is the number of days actually taken on average. Where the dark bar is taller than the pale one, that method typically misses its promise.

E. WHAT DOES THE RESULT TELL US?

Two things, immediately. First, almost all of the predictable delay risk in this dataset comes from the shipping method — everything else adds very little. Second, delivery methods differ far more in what they promise than in how fast they actually move goods.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It focuses attention. Rather than launching investigations into regional performance or supplier quality, this page says: the lever that matters is the shipping method and the promise attached to it. That is a much cheaper and faster thing to act on.

G. IMPORTANT THINGS TO REMEMBER

- The 54.8% late rate is measured against the company's own promised date, not an industry standard.
- The average profit figure is lower than the median because loss-making orders drag it down — both are shown so you can see this.
- The "high-risk orders" figure depends on where the cut-off is set. It uses 50% here; you can move that cut-off on the Model Diagnostics tab and watch the number change.

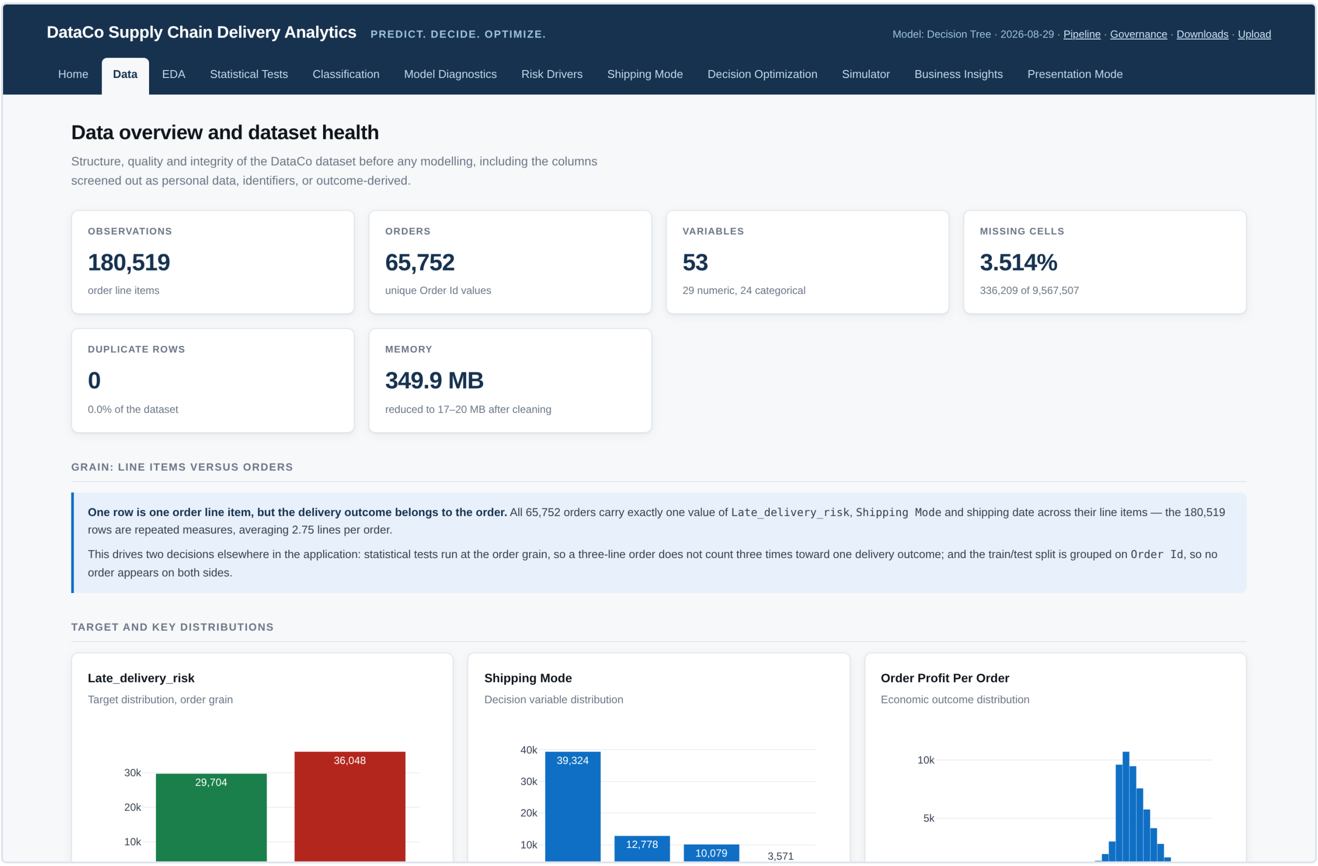
Data

A. WHAT IS THIS TAB?

This page shows exactly what information the analysis is built on, and how trustworthy it is. It counts the records, flags the missing values, points out columns that are empty or duplicated, and identifies the columns holding personal data.

B. WHY DO WE NEED THIS TAB?

Because a model is only as good as the data underneath it. Before anyone believes a prediction, they are entitled to ask: where did this come from, is any of it missing, and has anything been quietly thrown away? This page answers all three, in the open.



> **The Data tab.** Record counts and quality measures at the top, followed by the explanation of orders versus order lines.

C. WHAT DO WE SEE HERE?

- **Six count cards** — 180,519 observations, 65,752 orders, 53 variables, the percentage of missing values, the number of duplicate rows (zero), and how much computer memory the data occupies.
- **A "grain" explanation box** covering the difference between an order and an order line. This is the most important idea on the page.
- **Three distribution charts** — how many orders were late versus on time, how orders split across the four shipping methods, and the spread of profit.

- **A dictionary reconciliation section** comparing the actual data file against the accompanying description document, and listing any mismatches.
- **Sensitive and outcome-derived fields** — three lists of columns flagged as personal data, as unusable, or as identifiers.
- **An outlier table and a full column inventory** listing every column with its type, missing count and an example value.

D. HOW SHOULD I READ IT?

> **IN PLAIN ENGLISH: ORDERS VERSUS ORDER LINES**
>
> If a customer buys a shirt and a pair of shoes in one order, the file records that as two rows — one per product. But the delivery is a single event: both items arrive together, either on time or late.
>
> So counting rows would count that one delivery twice. Across the whole file, 180,519 rows describe just 65,752 actual deliveries — about 2.75 rows per delivery. The application therefore does its statistical work on the 65,752 orders, not the 180,519 rows.

In the **dictionary reconciliation** section, two genuine mismatches are reported. One column, `Order Zipcode`, exists in the data file but is not documented in the description file — and it is 86% empty, so it is excluded. Separately, the data file spells one column with a lower-case "s" (`shipping date (DateOrders)`) while the description file uses a capital. Small, but enough to break code that looks for the wrong spelling.

In the **columns carrying no signal** table, four columns are listed as useless: `Product Description` is entirely empty, `Customer Email` and `Customer Password` contain the same placeholder text in every row, and `Product Status` is the same value throughout. A column that never changes cannot explain anything that does change.

E. WHAT DOES THE RESULT TELL US?

The data is in good condition: no duplicate rows, very few missing values apart from the one 86%-empty column, and a clear unique reference for every order line. The issues found are all structural — empty columns, duplicated columns, a naming mismatch — rather than signs of corrupted or unreliable records.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It establishes that the conclusions rest on complete and consistent records, and it documents every exclusion. If someone challenges a finding by asking "did you leave something out?", this page is the answer — including the full inventory of all 53 columns.

G. IMPORTANT THINGS TO REMEMBER

- Personal data — customer names, street addresses and map coordinates — is present in the source file but excluded from every model. Handle the underlying file accordingly.
- Outliers were kept, not deleted. Very large orders and loss-making orders are real business events, not data errors.

- The final year of data covers January 2018 only. Any year-on-year comparison will show a misleading drop at the end, which the EDA tab flags explicitly.

EDA

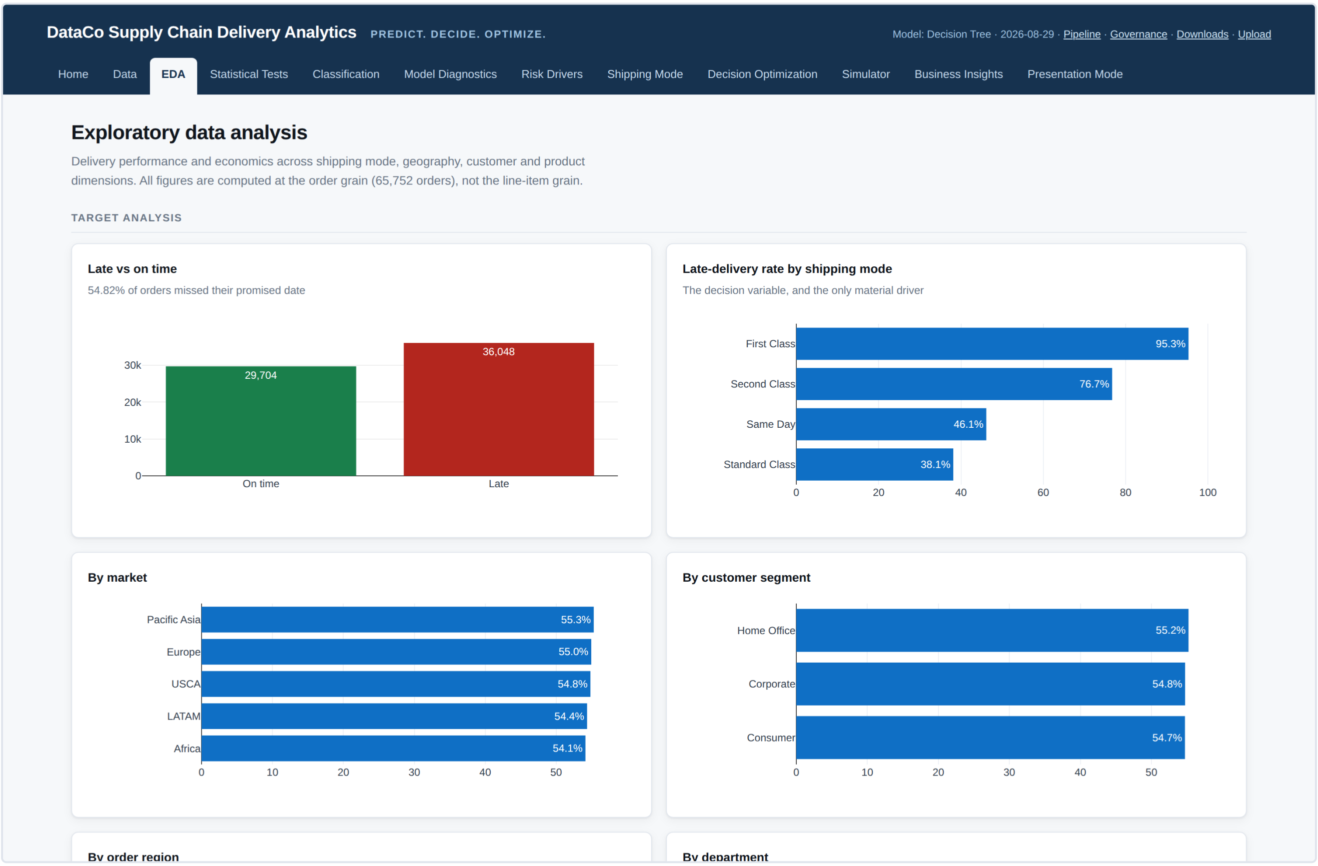
EDA stands for Exploratory Data Analysis — looking for patterns before building anything.

A. WHAT IS THIS TAB?

This is where we look at the data through charts, before any modelling. It compares late delivery rates across shipping methods, markets, regions, customer types and product categories, then does the same for profit.

B. WHY DO WE NEED THIS TAB?

To find out where the delays actually are. It answers: do some markets run later than others? Do expensive orders arrive late more often? Is the problem getting better or worse? It also gives us something to check the model against later — if the model says one thing and the charts say another, something is wrong.



> **The EDA tab.** Late-delivery rates broken down by shipping mode, market, customer segment and other dimensions.

C. WHAT DO WE SEE HERE?

- **Target analysis** — six charts showing the late rate broken down by shipping mode, market, customer segment, order region and department.
- **A mechanism section** comparing promised days against actual days for each shipping method, with a supporting table.

- **Financial analysis** — the spread of profit, average profit by shipping method, and profit compared between late and on-time orders.
- **Two interactive explorers** with drop-down menus: one for numerical variables (13 options) and one for categories (10 options).
- **A monthly trend chart** with a warning box about the partial final year.
- **Correlation analysis** — a coloured grid showing which numerical variables move together.

D. HOW SHOULD I READ IT?

The bar charts. Each bar is one category, and its length is the percentage of orders in that category that arrived late. A longer bar means more late deliveries. Look for bars that are noticeably different from the others.

> **THE MOST IMPORTANT THING TO NOTICE**
>
> On the market, region, segment and category charts, all the bars are **almost exactly the same length**. Across the five markets the late rate runs from 54.1% to 55.3% — a spread of about one percentage point.
>
> That flatness is the finding, not a boring chart. It says geography and product type do not explain delivery delays here. By contrast, the shipping-mode chart runs from 38.1% to 95.3% — a 57-point spread.

The correlation grid. Blue squares mean two measures rise together, red squares mean one rises as the other falls, and pale squares mean no relationship. It is useful for spotting variables that duplicate each other. The page carries an explicit warning that correlation shows association, not cause.

The monthly trend. Orders and late rate over time. The sharp fall at the right-hand edge is not a business improvement — it is where the data stops, in January 2018.

E. WHAT DOES THE RESULT TELL US?

Delivery delay in this dataset is concentrated almost entirely in the choice of shipping method. The transit-time table shows why: Second Class and Standard Class take essentially the same time to deliver (4.00 and 3.99 days on average), but Second Class promises two days where Standard Class promises four. Same journey, tighter promise, far more broken promises.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It rules things out, which is valuable. There is no point launching regional service improvement programmes when the regional differences are within a percentage point of each other. The evidence points at the promise, not the operation.

G. IMPORTANT THINGS TO REMEMBER

- These charts describe what happened. They do not prove that changing anything would produce a different result.
- All figures are per order, not per order line, so a large basket does not count several times.
- Ignore the final points on the monthly chart — January 2018 is a partial month of data.

TAB 4 OF 12

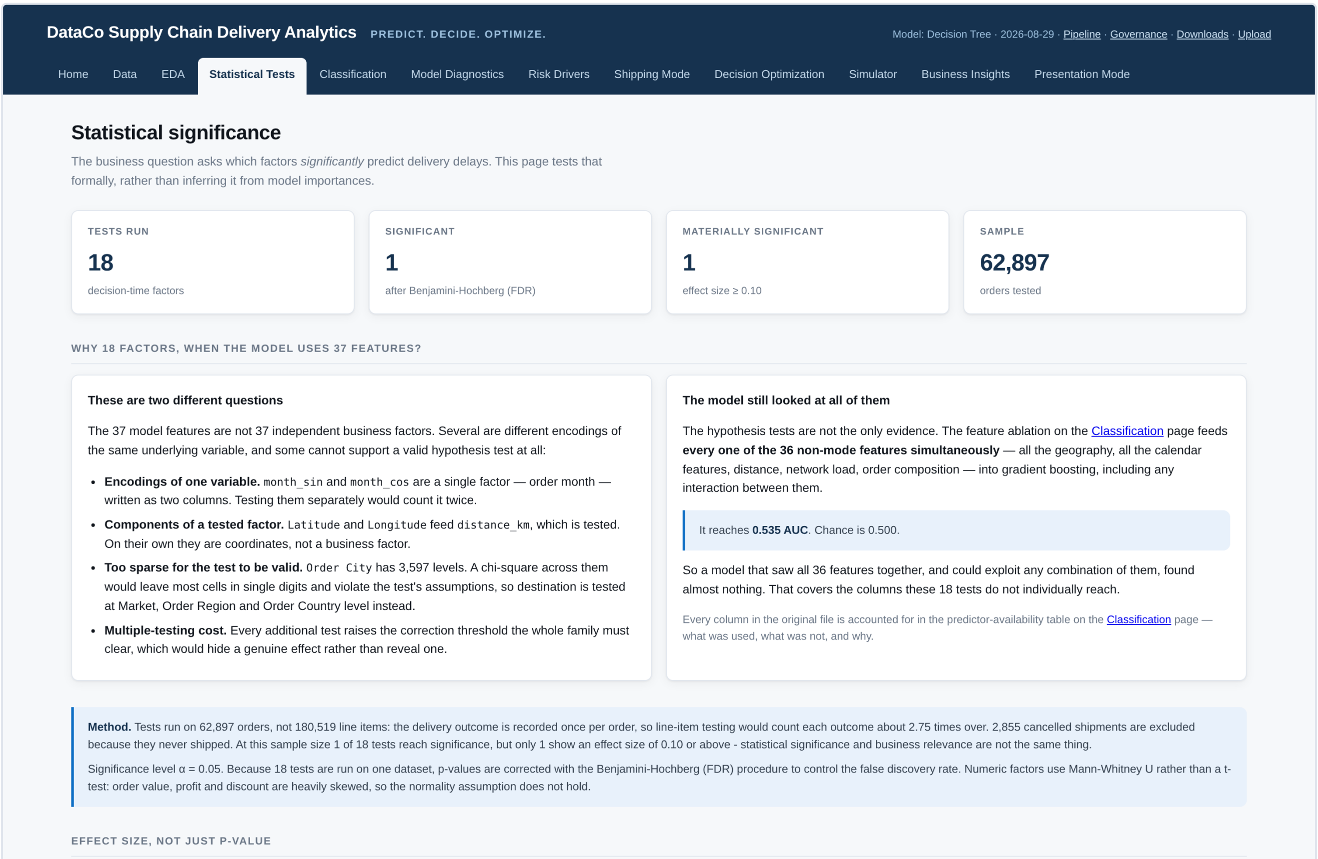
Statistical Tests

A. WHAT IS THIS TAB?

The previous tab showed patterns in charts. This one checks, formally, whether those patterns are real differences or just the kind of variation you would expect by chance. It runs 18 tests, one for each factor that might plausibly affect delivery.

B. WHY DO WE NEED THIS TAB?

Because eyes deceive. If you split any dataset into groups, the groups will differ a little, purely by luck. A statistical test tells you whether a difference is bigger than luck would explain. The business question specifically asks which factors significantly predict delays, so this tab answers it directly.



> **The Statistical Tests tab.** Summary counts at the top, then the explanation of why 18 factors are tested when the model uses 37 features.

C. WHAT DO WE SEE HERE?

- **Four summary cards** — 18 tests run, 1 significant, 1 materially significant, and 62,897 orders tested.
- **An explanation section** answering why 18 factors are tested when the model uses 37 features.
- **A method box** describing which test was used and why.
- **An effect-size chart** ranking all 18 factors by how much they actually matter.

- **The full results table** with, for each factor, the test used, the p-value, the effect size, whether it is significant, and a plain-English interpretation.
- **A hypotheses table** stating what each test was checking.
- **A separate mechanism test**, reported apart from the main set.

D. HOW SHOULD I READ IT?

> **IN PLAIN ENGLISH: P-VALUE AND EFFECT SIZE**
>
> A **p-value** answers "could this difference be a fluke?" A small p-value (below 0.05) means the difference is probably real. It says nothing about how big the difference is.
>
> An **effect size** answers the question that actually matters to a business: "is the difference big enough to care about?" Values near 0 mean the factor barely moves the needle; values above about 0.2 mean it matters a lot.
>
> You need both. With 62,897 orders, a test can detect differences far too small to act on — so a factor can be statistically significant and commercially irrelevant at the same time.

Read the results table by looking at the **effect size column first**, not the p-value. The table is deliberately sorted that way.

E. WHAT DOES THE RESULT TELL US?

Of the 18 factors tested, exactly one is statistically significant after correction: **Shipping Mode**, with an effect size of 0.481 — classed as large. Everything else — market, region, country, product category, department, customer segment, order value, quantity, discount, distance and how busy the network was — fails to reach significance.

This is a stronger result than it first appears. With nearly 63,000 orders, these tests have enormous power to detect even small effects. Failing to find one means the effect is essentially zero, not merely small.

> **WHY 18 TESTS WHEN THE MODEL USES 37 FEATURES?**
>
> The 37 model features are not 37 separate business factors. Some are two columns describing one thing (the calendar month is stored as a pair of values). Some are ingredients of a factor that is tested — map coordinates feed into the distance measure. And some have too many categories for a valid test: `Order City` has 3,597 different values, which would leave most groups with only a handful of orders.
>
> The application answers the "did you miss something?" question a different way. The Classification tab feeds all 36 non-shipping-mode features at once into a model, including any combinations between them. That model scores 0.535, where 0.500 is a coin toss.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It prevents wasted effort. Without this tab, someone would look at the EDA charts, spot that one market is 1.2 points worse than another, and commission a project about it. This tab shows that difference is indistinguishable from noise.

G. IMPORTANT THINGS TO REMEMBER

- Significant does not mean important. The table gives both measures for exactly this reason.
- 2,855 cancelled orders are excluded from these tests. They never shipped, so calling them "delivered on time" would be misleading.
- Because 18 tests are run at once, the results are adjusted so that running many tests does not throw up false positives by chance.
- These results describe this dataset. They are not a general finding that geography never affects delivery in the real world.

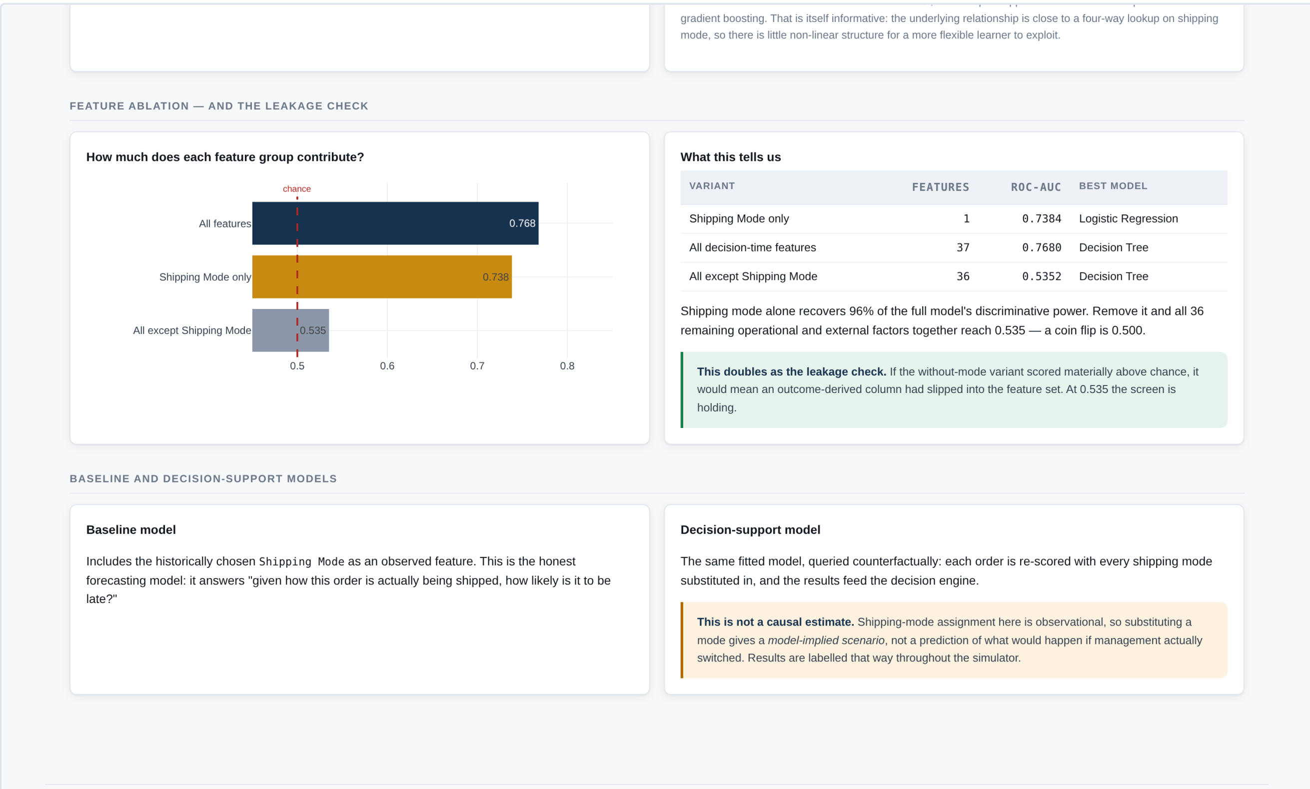
Classification

A. WHAT IS THIS TAB?

This is where the prediction model is built. "Classification" simply means sorting things into categories — here, sorting orders into "likely to be late" and "likely to be on time". The page shows which information the model is allowed to use, how the data was split for testing, and how four different modelling approaches compared.

B. WHY DO WE NEED THIS TAB?

It answers: given only what we know at the moment we choose a shipping method, how well can we predict whether this order will be late? Everything downstream depends on this prediction being honest, and this page is where honesty is demonstrated.



The Classification tab. The feature-ablation section, showing how much predictive power comes from shipping mode alone.

C. WHAT DO WE SEE HERE?

- **Predictor availability and data leakage** — a warning box plus a 54-row table listing every column and whether it was used.
- **Train/test split cards** — 138,493 rows for training, 34,272 held back for testing, 37 predictors, and the balance between late and on-time orders.
- **A model comparison table** with eight scores for each of four models.
- **A feature ablation section** — a chart and table showing what happens when features are removed.

- **Baseline and decision-support model panels** explaining the two ways the same model is used.

D. HOW SHOULD I READ IT?

> **IN PLAIN ENGLISH: DATA LEAKAGE**
>
> A model should only use information that would genuinely be known at the moment the decision is made. If you are predicting whether an order will be late before it ships, you cannot use information that only exists after it has been delivered.
>
> Think of predicting tomorrow's weather. Using today's air pressure is fair. Using tomorrow's rainfall reading is not a prediction — it is looking at the answer.

Three columns in this dataset are exactly that kind of answer-peeking, and all three are excluded:

COLUMN	WHY IT WAS EXCLUDED
`Delivery Status`	Literally records whether the delivery was late. Matches the answer on 100% of rows.
`Days for shipping (real)`	How long shipping actually took — known only after the fact. Reproduces the answer on 97.6% of rows.
`shipping date (DateOrders)`	Subtract the order date from it and you recover the actual transit time.

> **WATCH OUT FOR**
>
> Including any of these produces accuracy figures around 99%, which is what most published analyses of this dataset report. That number is not a forecast — it is the model reading the answer off the page. This application's lower score is the honest one.

The split cards. The model is trained on 138,493 rows and then tested on 34,272 it has never seen, so the scores reflect performance on genuinely new orders. The split keeps all lines of the same order together, so an order can never appear on both sides.

The model comparison table. Four modelling approaches, eight scores each. Each score is explained fully on the next tab. The preferred model is highlighted:

MODEL	ROC-AUC	ACCURACY	F1	NOTES
Decision Tree — preferred	0.7680	0.7172	0.7049	Best separation
Gradient Boosting	0.7653	0.7074	0.7046	Most sophisticated
Random Forest	0.7598	0.7184	0.7044	Many trees combined
Logistic Regression	0.7423	0.7010	0.6852	Simplest, most explainable

All four land within 0.03 of each other, and a simple decision tree beats the most sophisticated method. That is informative: it means the underlying relationship is close to a simple four-way lookup on shipping mode, with no hidden complexity for a cleverer model to find.

The ablation chart. "Ablation" means removing something to see what happens. Three bars:

WHAT THE MODEL CAN USE	SCORE	READING
Shipping mode only	0.7384	Nearly all the predictive power
All 37 features	0.7680	Everything adds a little more
All 36 features except shipping mode	0.5352	Barely better than a coin toss

E. WHAT DOES THE RESULT TELL US?

The model can meaningfully separate risky orders from safe ones, and virtually all of that ability comes from the shipping method. The third bar also serves as a safety check: if the model scored well without shipping mode, that would suggest some other outcome-revealing column had slipped in. At 0.535, the screening is holding.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It tells you the prediction is trustworthy — built only on information available when the decision is made — and that the factor driving it is one you control. A prediction based on something you cannot change would be interesting but useless.

G. IMPORTANT THINGS TO REMEMBER

- A score of 0.768 is genuinely useful, not weak. Compare it against 0.500 for guessing, not against 100%.
- The "decision-support" use of the model — re-scoring an order under each shipping method — produces a model-implied scenario, not a guarantee of what would happen.
- 7,754 cancelled order lines are excluded from training. They never shipped.

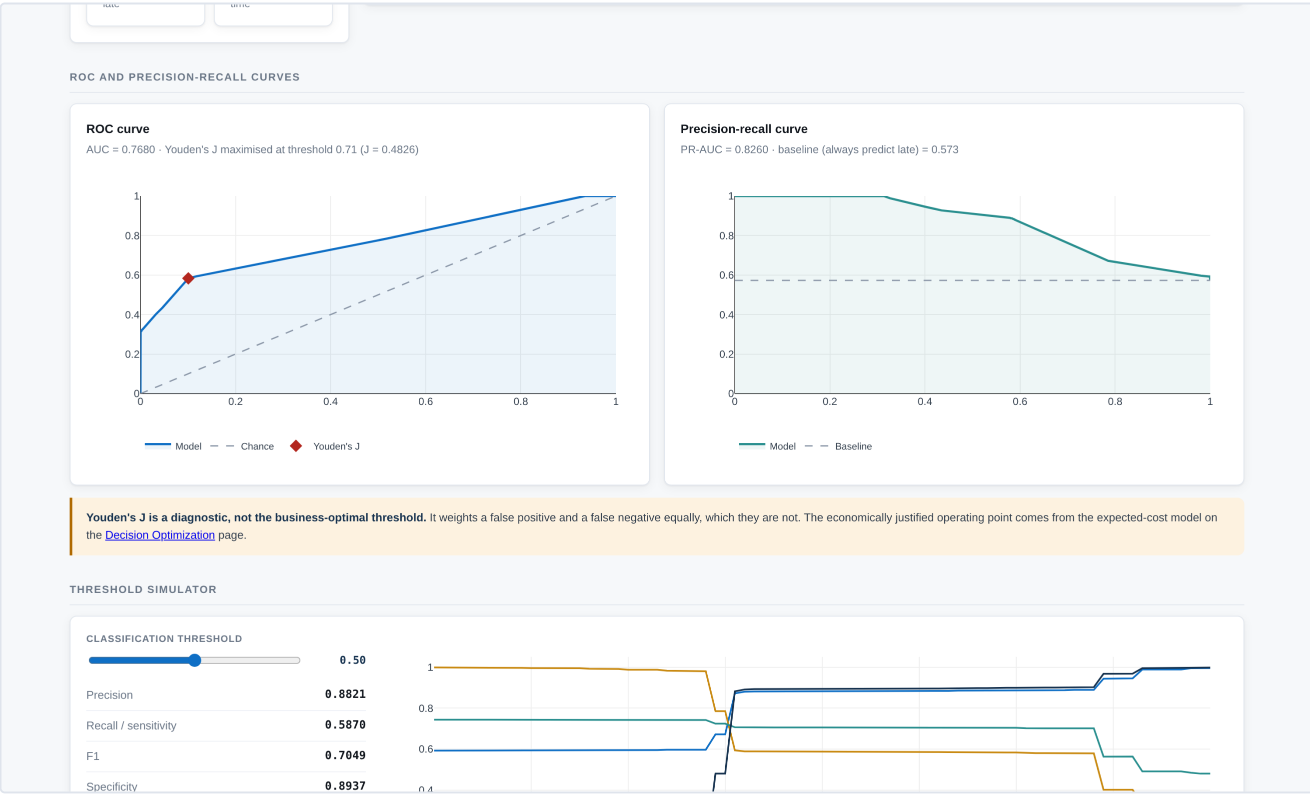
Model Diagnostics

A. WHAT IS THIS TAB?

The previous tab said which model was chosen. This one examines how good it actually is, in detail: what kinds of mistakes it makes, whether its probabilities can be trusted, and what happens if you make it more or less cautious.

B. WHY DO WE NEED THIS TAB?

Because a single accuracy figure hides the important question: what kind of mistake does it make, and which mistake costs more? Missing a late order and raising a false alarm are not equally expensive, and this page separates them.



The Model Diagnostics tab. ROC and precision-recall curves, with the interactive threshold simulator below.

C. WHAT DO WE SEE HERE?

- **A confusion matrix** — four cards counting each type of correct and incorrect prediction, with a table explaining what each costs.
- **Two curves** — ROC and precision-recall.
- **A threshold simulator** — a slider that updates eight measures live.
- **Logistic regression diagnostics** — eight statistical measures plus a table of odds ratios.
- **A multicollinearity (VIF) table.**
- **A calibration section** — a curve and a table.

D. HOW SHOULD I READ IT?

IN PLAIN ENGLISH: THE FOUR BOXES

Every prediction falls into one of four boxes. Using the actual figures:

- **True positive (11,577)** — predicted late, and it was late. Correct, and actionable: we can upgrade the shipping or warn the customer.
- **True negative (13,003)** — predicted on time, and it was. Correct, no action needed.
- **False positive (1,547)** — predicted late, but it arrived on time. A false alarm. We may have paid for a needless upgrade. Annoying, but the cost is known in advance and bounded.
- **False negative (8,145)** — predicted on time, but it was late. The expensive mistake: no warning was given, nothing was done, and the customer is surprised.

The model is currently cautious about crying wolf: it is right 88% of the time when it says "late" (precision 0.8821), but it only catches 59% of the orders that actually are late (recall 0.5870). For a service business, where the unflagged late delivery is the costly one, you would often want to catch more of them and accept more false alarms. The slider lets you do exactly that.

MEASURE	VALUE	PLAIN MEANING
Accuracy	0.7172	Share of all predictions that were correct. Misleading on its own.
Precision	0.8821	When it says "late", how often is it right?
Recall / Sensitivity	0.5870	Of all the genuinely late orders, how many did it catch?
F1 score	0.7049	A single number balancing precision and recall.
ROC-AUC	0.7680	How well it separates risky from safe orders. 0.5 is guessing.
PR-AUC	0.8260	Similar, but focused on how well it finds the late orders specifically.
Brier score	0.1776	How accurate its probabilities are. Lower is better.
Youden's J	0.4826	A balance measure, best at a cut-off of 0.71.

The two curves. On the ROC curve, the line bows toward the top-left corner; the further from the diagonal dashed line, the better. The dashed line is what pure guessing would look like. On the precision-recall curve, the trade-off is visible directly: catching more late orders (moving right) means more false alarms (the line falls).

The threshold slider. The model outputs a probability, and the threshold is the cut-off at which you call an order "risky". Lower it to catch more late orders at the cost of more false alarms; raise it to act only on the clearest cases. Drag it and all eight measures update instantly.

WATCH OUT FOR

The page reports Youden's J, which suggests a cut-off of 0.71. That measure treats a false alarm and a missed late order as equally bad — which they are not. The economically justified cut-off comes from the cost model on the Decision Optimization tab, not from this statistic.

Odds ratios. An odds ratio above 1 means higher odds of late delivery than the reference category; below 1 means lower. So an odds ratio of 2.0 for a shipping method means roughly double the odds of lateness compared with the baseline method.

VIF (multicollinearity). This checks whether two predictors are telling the model the same thing. Four predictors are flagged here, and that is expected: the promised delivery window is completely determined by the shipping method, so the two are effectively the same variable. The application keeps only shipping mode in the predictive model for this reason.

Calibration. This asks whether the probabilities mean what they say — if the model says "70% likely to be late" for a group of orders, did about 70% of them arrive late? Points near the diagonal line mean yes. This matters because the decision engine multiplies these probabilities by money.

E. WHAT DOES THE RESULT TELL US?

The model is reliable enough to base decisions on, and its probabilities are reasonably well-calibrated. Its main weakness is that at the default cut-off it misses about four in ten genuinely late orders — a choice you can change with the slider.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It converts a technical question into a business one. Rather than asking "is the model good?", you can ask "how many late deliveries are we willing to miss in order to avoid a given number of unnecessary upgrades?" — and set the cut-off accordingly.

G. IMPORTANT THINGS TO REMEMBER

- All figures come from held-out data the model never saw during training.
- There is no single "correct" threshold. It depends on the relative cost of the two mistakes, which is a business judgement.
- Odds ratios describe association within the model, not proven cause.

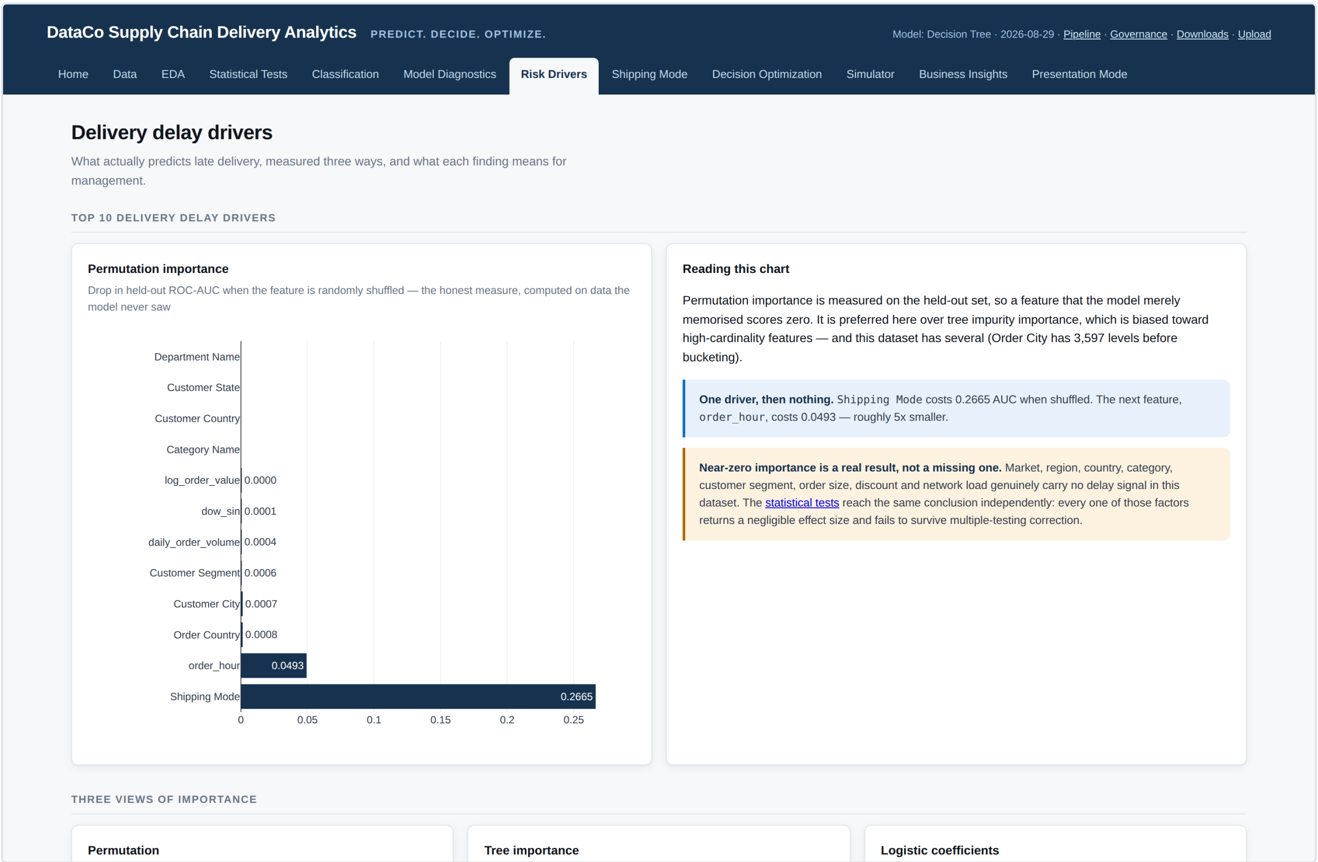
Risk Drivers

A. WHAT IS THIS TAB?

This page opens up the model and shows which pieces of information it actually relies on when predicting delay. It measures this three different ways, then translates each finding into business language with a suggested action.

B. WHY DO WE NEED THIS TAB?

A prediction nobody can explain is hard to act on and harder to defend. This tab answers why the model thinks an order is risky — which is what turns a score into something a manager can do something about.



The Risk Drivers tab. Permutation importance ranking, with the interpretation panel beside it.

C. WHAT DO WE SEE HERE?

- **A permutation importance chart** ranking the top factors.
- **Three smaller charts** comparing three different ways of measuring importance.
- **Business interpretation blocks** for the top ten drivers, each with a statistical reading, a business reading and a suggested action.
- **A four-panel section** distinguishing association, prediction, causation and decision scenario.

D. HOW SHOULD I READ IT?

> **IN PLAIN ENGLISH: PERMUTATION IMPORTANCE**
>
> To test whether a piece of information matters, scramble it — shuffle that column randomly so it becomes meaningless — and see how much worse the model gets. If performance collapses, that information was doing real work. If nothing changes, it was not.
>
> It is the equivalent of removing one ingredient from a recipe to find out whether anyone notices.

The actual ranking:

RANK	FACTOR	IMPORTANCE	READING
1	`Shipping Mode`	0.2665	By far the largest. Scrambling it damages the model badly.
2	`order_hour`	0.0493	Hour of day the order was placed. Five times smaller, but clearly above the rest.
3	`Order Country`	0.0008	Effectively zero.
4	`Customer City`	0.0007	Effectively zero.
5	`Customer Segment`	0.0006	Effectively zero.

The pattern is one dominant driver, one small secondary one, and then a cliff. Everything from third place down is within rounding distance of zero.

> **ABOUT THAT SECOND PLACE**
>
> `order_hour` is the hour of day the order was placed. It is worth being precise: it is not zero, but it is five times smaller than shipping mode. The most likely explanation is that it acts as a partial stand-in for the shipping method — if the mix of shipping methods varies by time of day, then knowing the hour tells the model a little about which method was used. It was not one of the 18 factors put through a formal statistical test.

Three measures compared. The three smaller charts use different methods, each with different blind spots — one favours factors with many categories, another only sees straight-line relationships. When all three agree, as they do here, the finding is a property of the data rather than a quirk of one technique.

E. WHAT DOES THE RESULT TELL US?

Delivery delay risk in this dataset is driven overwhelmingly by the shipping method chosen. Destination country, city, customer type, product category, order size and network load carry essentially no usable signal. This agrees with the Statistical Tests tab, which reached the same conclusion by a completely different route.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It says where to invest attention and, just as usefully, where not to. A near-zero importance is a real result: it means building destination-level or product-level delay forecasting would not pay for itself here.

G. IMPORTANT THINGS TO REMEMBER

- **An important predictor is not a proven cause.** Shipping mode predicts lateness; that does not prove changing it would change the outcome.
- The four panels at the bottom of the page set out this distinction explicitly: association, model prediction, causation, and decision scenario are four different things.
- Importance is measured on data the model never saw, so it cannot be inflated by the model simply memorising the training set.

TAB 8 OF 12

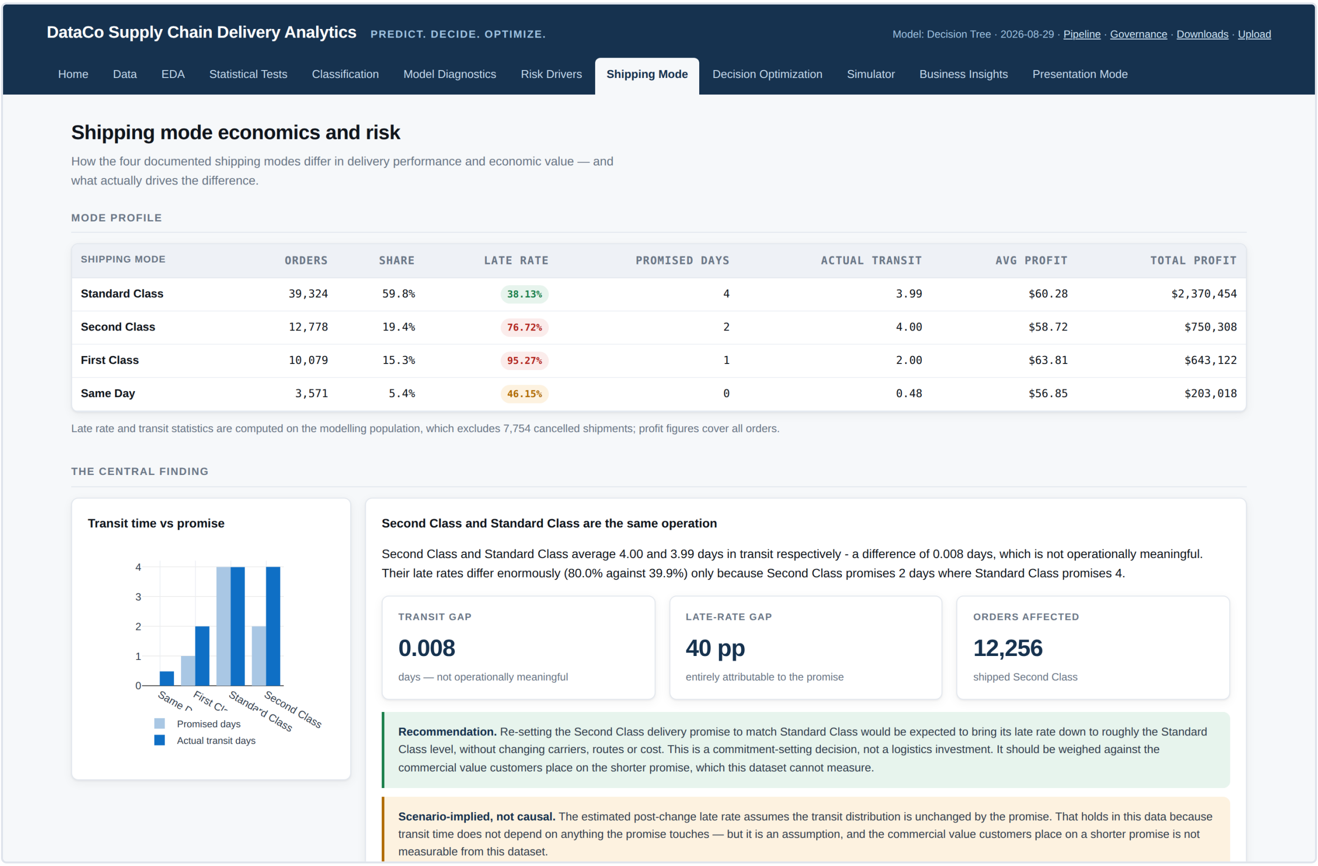
Shipping Mode

A. WHAT IS THIS TAB?

This page examines the one thing management can actually change. It compares all four shipping methods on delivery performance, speed and profitability, and explains what really drives the differences between them.

B. WHY DO WE NEED THIS TAB?

Everything so far has pointed here. If shipping mode is the dominant driver and the only controllable one, we need to understand precisely how the four options differ before we can decide between them.



The Shipping Mode tab. The four-mode comparison table, followed by the promised-versus-actual analysis.

C. WHAT DO WE SEE HERE?

- **A mode profile table** — for each method: order count, share, late rate, days promised, days actually taken, and profit.
- **The central finding section** — a chart and analysis of promise versus performance.
- **A trade-off chart** plotting late rate against speed.
- **A profit-by-mode chart.**
- **A segment comparison** with a drop-down for market, region, segment or department.
- **An economic outcome section** comparing profit on late versus on-time orders.

D. HOW SHOULD I READ IT?

The mode profile table is the heart of the page. The two columns to compare are promised days and actual transit:

SHIPPING METHOD	ORDERS	PROMISED	ACTUAL	LATE RATE
Standard Class	37,632	4 days	3.99 days	39.9%
Second Class	12,256	2 days	4.00 days	80.0%
First Class	9,602	1 day	2.00 days	100.0%
Same Day	3,407	0 days	0.48 days	48.4%

THE CENTRAL FINDING

Look at Standard Class and Second Class. They take **the same time** to deliver — 3.99 days versus 4.00 days, a difference of eight thousandths of a day. Yet Standard Class is late 39.9% of the time and Second Class 80.0% of the time.

The entire 40-point gap comes from the promise. Standard Class promises four days and takes four. Second Class promises two days and also takes four. Same operation, tighter promise, twice as many broken promises.

First Class is the extreme case: it promises one day and always takes exactly two, so it is late 100% of the time. It is not slow — it is the second-fastest method — it is simply over-promised.

The trade-off chart plots each method with speed on one axis and late rate on the other. Standard Class is the slowest but the most reliable against its own promise; Same Day is eight times faster but breaks its promise more often. Neither is best on both counts, which is exactly why a cost model is needed.

WATCH OUT FOR

Do not assume the method with the lowest late rate is automatically the best. Standard Class wins on reliability precisely because it promises the least. And a customer who wants goods tomorrow is not served by a four-day delivery that arrives "on time".

The economic section reports something important: profit on late orders and profit on on-time orders are almost identical. The correlation between lateness and profit is −0.0056, which is effectively zero.

E. WHAT DOES THE RESULT TELL US?

Two things. First, the delay problem here is a promise-setting problem, not a logistics problem — the goods move at the speed they move, and what varies is what was committed to. Second, because late delivery has no measurable effect on profit in this data, the cost of a late delivery cannot be calculated from the data and must be supplied as an assumption.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It surfaces a genuinely cheap intervention. Re-setting the Second Class promise to match Standard Class would be expected to bring its late rate down to roughly the Standard Class level, affecting 12,256 orders, without changing a single carrier, route or cost. That is a commitment decision, not a logistics investment.

G. IMPORTANT THINGS TO REMEMBER

- The estimated effect of re-promising assumes transit times stay the same. That holds in this data, but it is an assumption.
- A shorter promise has commercial value to customers that this dataset cannot measure. Lengthening it is not free in business terms, even though it is free operationally.
- Levels with fewer than 200 orders are excluded from the segment chart, because small samples produce unreliable percentages.
- Profit differences between modes are descriptive. They do not prove that a mode causes higher or lower profit.

Decision Optimization

A. WHAT IS THIS TAB?

This is where prediction becomes decision. It puts a money figure on each shipping option for an order, adds up the total expected cost of each, and recommends the cheapest. You set the cost assumptions yourself with sliders, and everything recalculates as you move them.

B. WHY DO WE NEED THIS TAB?

Knowing an order is likely to be late does not tell you what to do. Upgrading costs money; not upgrading risks an unhappy customer. This page prices both sides so the choice can be made on evidence rather than instinct.

> **SCENARIO ASSUMPTIONS — NOT OBSERVED VALUES**
>
> **The dataset contains no shipping-cost information whatsoever.** There are no freight charges, no carrier invoices, no logistics-cost fields of any kind. Total logistics cost therefore cannot be measured from this data.
>
> Every money figure on this page is an assumption you set, and the application labels it as such wherever it appears. This is deliberate: inventing plausible-looking shipping costs and presenting them as findings would be dishonest. Instead the page shows you how the answer changes as the assumptions change.

SHIPPING-MODE DECISION ENGINE

Expected cost by mode, at the median order value ($164.97)

Averaged across the scored held-out sample of 5,000 orders

SHIPPING MODE	MEAN P(LATE)	FREIGHT	EXPECTED DELAY COST	HOLDING	SPEED COST	EXPECTED TOTAL	MEAN TRANSIT	RECOMMENDATION
Standard Class	39.6%	$6.00	$9.20	$0.33	$31.96	**$47.49**	3.99 d	Lowest expected cost
Same Day	50.7%	$32.00	$11.79	$0.04	$3.87	**$47.70**	0.48 d	+$0.21 vs best
First Class	100.0%	$18.00	$23.25	$0.16	$16.00	**$57.41**	2.00 d	+$9.93 vs best
Second Class	80.2%	$10.00	$18.64	$0.33	$32.02	**$60.99**	4.00 d	+$13.50 vs best

LOGISTICS NETWORK POLICY COMPARISON

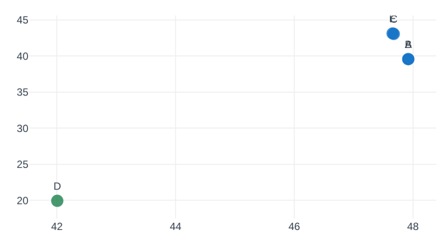
Scope. These are logistics SERVICE-network policies - which shipping mode is used for which orders. The dataset carries no warehouse, distribution-centre, facility or carrier fields, and origin geography has only two values, so it cannot support physical network design such as facility location.

Policies compared

Ranked by expected total cost under the current assumptions

POLICY	AVG COST/ORDER	EXPECTED LATE RATE	SAVING	COMPLEXITY
D - Risk-based assignment	$42.01	20.0%	12.3%	Highest - a per-order scoring decision, needs the model in the fulfilment path.
E - Risk + segment policy	$47.66	43.1%	0.6%	Moderate - one rule per segment and value band.
C - Region-specific policy	$47.68	43.1%	0.5%	Low - one rule per market or region.

Cost vs delivery reliability



The Decision Optimization tab. The shipping-mode decision engine and the policy comparison, both recalculating live as the assumption sliders move.

C. WHAT DO WE SEE HERE?

- **An assumptions panel** on the left with sliders for freight cost per method, the cost of a late delivery, holding cost, and the value of faster delivery — plus three preset buttons (low, medium and high delay cost).
- **The cost formula**, written out with each component colour-coded as either a model output or an assumption.
- **A degeneracy demonstration** — two charts showing what happens with and without the time-related cost terms.
- **The decision engine table** — expected cost broken into components for all four methods.
- **A policy comparison** of five different shipping strategies.
- **Break-even analysis** for each upgrade step.
- **Three sensitivity charts.**
- **A risk/value matrix** sorting orders into four action groups.

D. HOW SHOULD I READ IT?

> **IN PLAIN ENGLISH: EXPECTED COST**
>
> "Expected cost" means the average cost you would pay if you made the same decision many times. If a late delivery costs £100 and there is a 30% chance of being late, the expected cost of lateness is £30 — not because you ever pay exactly £30, but because that is what it averages out to across many orders.

The formula adds four things together for each shipping method:

COMPONENT	WHAT IT REPRESENTS
Freight cost	What that method costs to buy. Your assumption.
Expected delay cost	The chance of being late (from the model) multiplied by what a late delivery costs. Model output × your assumption.
Holding cost	The cost of capital tied up in goods while they are in transit. Your assumption.
Speed cost	The business value given up for each extra day in transit. Your assumption.

WHY THE LAST TWO COMPONENTS EXIST

A natural way to write this would be just freight + chance of late × cost of late. On this dataset that produces a useless answer. Standard Class has both the lowest assumed freight cost and the lowest late rate, so it wins for every single order no matter what numbers you put in — the optimisation answers itself before it starts. The page demonstrates this: with the time terms removed, Standard Class takes **99.8%** of orders.

What that simple formula misses is that faster methods genuinely deliver sooner even when they miss their promise. Same Day averages half a day in transit against Standard Class's four days. The holding and speed terms put a value on that, which restores a real trade-off — one that depends on the individual order.

You can see this for yourself: set both time sliders to zero and watch the recommendation collapse onto a single method.

The decision engine table shows each method with its cost broken into those four parts, plus the total. The cheapest is highlighted, and the others show how much more they would cost.

The policy comparison tests five ways of running the shipping operation. Under the default assumptions:

POLICY	COST PER ORDER	EXPECTED LATE RATE	SAVING
D — Risk-based assignment	$42.01	20.0%	12.35%
E — Risk + segment policy	$47.66	43.1%	0.55%
C — Region-specific policy	$47.68	43.1%	0.51%
A — Single mode for everything	$47.92	39.6%	–
B — Market-specific policy	$47.92	39.6%	0.00%

Policy D — choosing per order based on its risk and value — is both the cheapest and the most reliable. Note that market- and region-specific policies barely help, which is consistent with everything the earlier tabs found about geography.

Break-even analysis answers: how likely would a delay have to be before upgrading pays for itself? If the actual chance of delay is above the break-even point, the upgrade is justified under your assumptions.

Sensitivity charts vary one assumption at a time across a wide range and mark the point where the recommendation flips. Where no flip occurs, the recommendation is robust to that assumption.

The risk/value matrix plots orders by predicted risk against economic exposure, splitting them into four groups: Prioritise (high risk, high value), Watch/manage (high risk, low value), Protect (low risk, high value) and Routine (low risk, low value).

ON "LOGISTICS NETWORK DESIGN"

The business question mentions network design. This dataset cannot support the classic version of that — deciding where to build warehouses — because it contains no warehouse, distribution-centre, facility or carrier information, and only two origin locations. The page states this explicitly and scopes the decision to service-network design: which shipping method is used for which orders.

E. WHAT DOES THE RESULT TELL US?

Under the default assumptions, assigning shipping method order by order, based on predicted risk and order value, beats every fixed rule — about 12% cheaper than using one method for everything, while roughly halving the expected late rate. The sensitivity charts show which assumptions that conclusion depends on most.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It turns an argument into a calculation. Instead of debating whether faster shipping is "worth it", you state what you believe a late delivery costs, and the page tells you what follows. If colleagues disagree about the numbers, you can test both views in a minute.

G. IMPORTANT THINGS TO REMEMBER

- Every currency figure is an assumption, not a measurement. Change the sliders and every number changes.
- The savings are modelled estimates under those assumptions, not a guarantee.
- The model gives the probability of lateness; the assumptions give it a price. Both are needed, and only the first comes from data.
- Policy D requires the model to run inside the fulfilment process. That is an operational cost the comparison does not include.

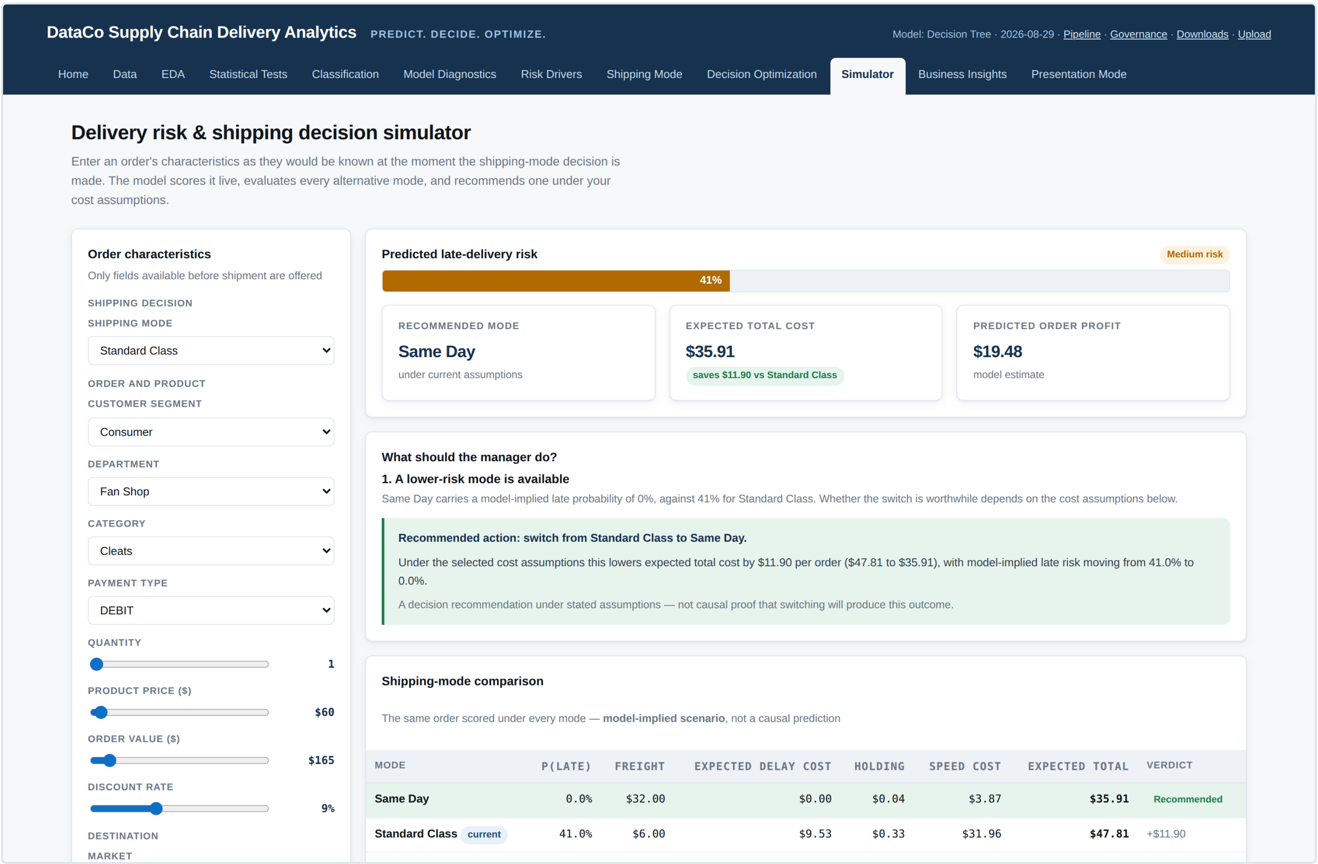
Simulator

A. WHAT IS THIS TAB?

An interactive tool for a single order. You describe an order using drop-downs and sliders, and the application predicts its delay risk live, evaluates all four shipping methods for it, and recommends one — with a plain-English explanation of why.

B. WHY DO WE NEED THIS TAB?

Because managers think in terms of specific orders, not aggregate statistics. This is where the whole analysis becomes usable: this order, going there, worth this much — what should we do with it?



The Simulator tab. Order inputs on the left; live risk meter, recommendation and manager guidance on the right.

C. WHAT DO WE SEE HERE?

- **An order characteristics panel** — shipping method, customer segment, department, category, payment type, quantity, product price, order value, discount rate, and destination market, region and country.
- **A delivery promise control**, locked to the shipping method by default with an override checkbox.
- **A cost assumptions panel** with the same sliders and presets as Tab 9.
- **A risk meter** — a coloured bar showing the predicted probability, with a Low/Medium/High label.

- **Three result cards** — recommended method, expected total cost, and predicted order profit.
- **A "What should the manager do?" panel** listing the specific risk factors for this order and a recommended action.
- **A shipping-mode comparison table** and two charts.
- **A what-if analysis tool** at the bottom.

D. HOW SHOULD I READ IT?

Only fields that would genuinely be known before shipping are offered as inputs. You will not find "actual delivery date" here, because that would not exist at decision time.

The risk meter fills from left to right and changes colour: green for low risk, amber for medium, red for high. The percentage is the model's estimated probability that this order arrives after its promised date.

The comparison table shows all four methods for this same order, with cost broken into components. The recommended row is highlighted; the others show how much extra they would cost.

> **THE PROMISE OVERRIDE — THE MOST INTERESTING CONTROL**
>
> By default the promised delivery window follows the shipping method automatically, because in this dataset the two are locked together. Ticking "override the promise" unlocks it.
>
> That models something quite different from changing the shipping method: it models changing the commitment while leaving the operation untouched. It is the interactive version of the central finding on Tab 8 — that Second Class and Standard Class are the same operation with different promises.

What-if analysis. Pick a variable, choose a different value, and press "Run scenario". The result table shows the predicted risk before and after, the change in percentage points, the change in expected cost, and whether the recommendation changes.

> **MODEL-IMPLIED SCENARIO, NOT A GUARANTEED OUTCOME**
>
> The application labels every what-if result as a "model-implied scenario change", and that wording matters. Changing a value and re-running tells you what the model expects for an order with those characteristics. It does **not** tell you what would happen if you actually made that change in the real world.
>
> The difference: the model has learned that orders shipped a certain way tend to be late. It has not learned that shipping a particular order that way would make it late. Only a controlled experiment could establish that.

E. WHAT DOES THE RESULT TELL US?

For any order you describe, it gives a risk estimate, a recommended shipping method under your assumptions, and the reasoning behind it. Experimenting shows quickly that changing the shipping method moves the prediction substantially, while changing the market or category moves it barely at all — the headline finding, made tangible.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It is the tool a planner would actually use. It also makes the analysis persuasive: rather than describing the finding that geography does not matter, you can change the destination in front of an audience and let them watch the number stay put.

G. IMPORTANT THINGS TO REMEMBER

- Results are model-implied scenarios, not guarantees.
- Fields you leave alone are set to typical values from the data, so the order is always complete and realistic.
- Unusual combinations may fall outside anything in the historical data, where the model is less reliable.
- The cost figures depend on the assumption sliders, exactly as on Tab 9.

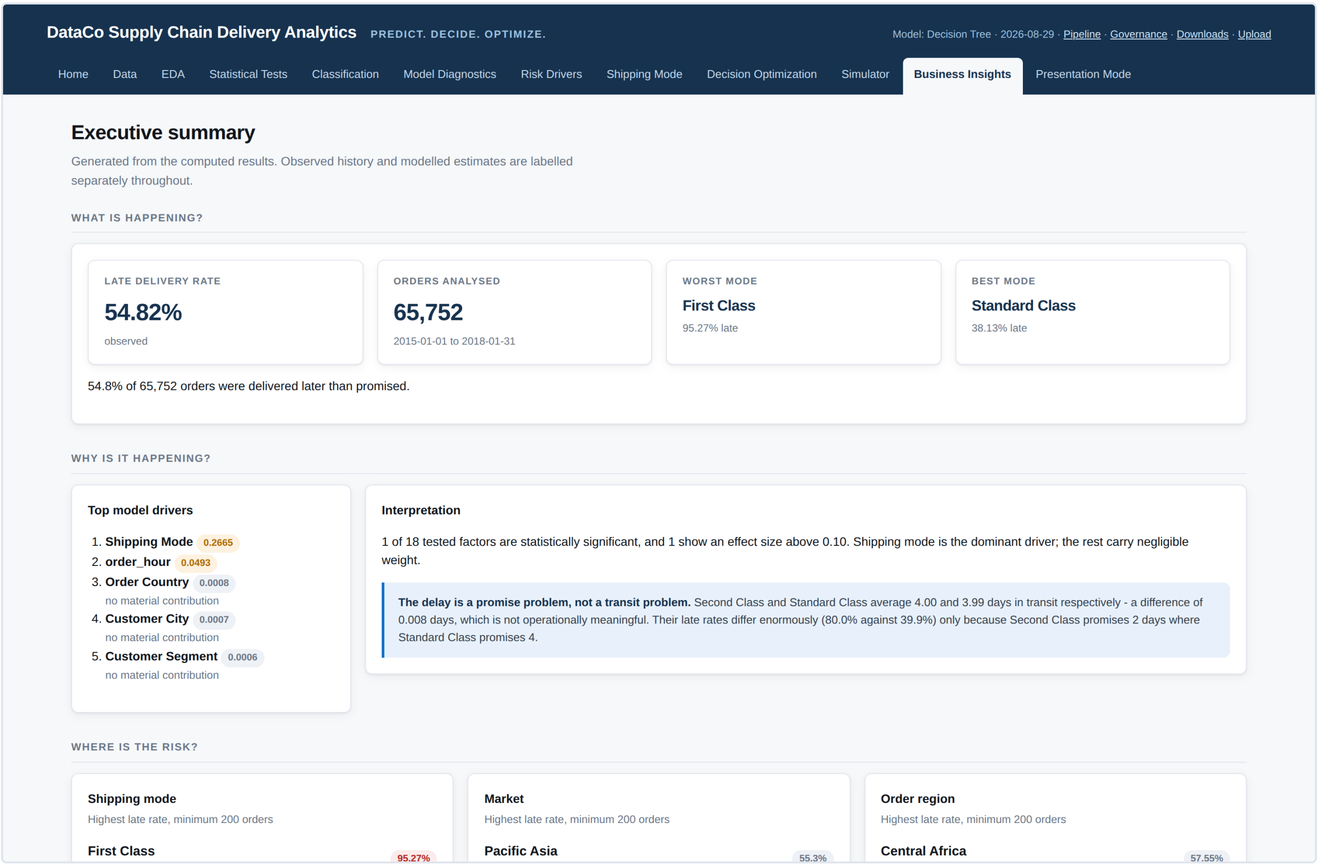
Business Insights

A. WHAT IS THIS TAB?

The executive summary. It pulls the findings from every other tab into one page structured around five questions: what is happening, why, where the risk is, what management should do, and what it is worth.

B. WHY DO WE NEED THIS TAB?

Because a senior audience will not click through twelve tabs. This page is what you would put in front of a director who has five minutes — and it is generated from the computed results, not written by hand, so it cannot drift out of line with the analysis.



> **The Business Insights tab.** The executive summary, structured as what is happening, why, and where the risk sits.

C. WHAT DO WE SEE HERE?

- **"What is happening?"** — headline figures with a one-line summary.
- **"Why is it happening?"** — the top model drivers, each marked as material or not.
- **"Where is the risk?"** — the highest-risk shipping method, market, region, category and customer segment.
- **Segment dashboards** — four charts pairing late rate with average profit.
- **"What should management do?"** — four recommendations, each tagged with the kind of evidence behind it.

- **"What is the economic implication?"** — the assumptions and the modelled impact.

D. HOW SHOULD I READ IT?

Read the tags on each recommendation carefully. They are not decoration — they tell you what kind of claim is being made:

TAG	WHAT IT MEANS
Observed transit distributions	Measured directly from the data. The strongest kind of evidence here.
Held-out model performance	Based on how the model performed on data it had never seen.
Modelled estimate under assumptions	Depends on the cost assumptions. Change them and this changes.
Observed historical rates	What actually happened, described — not a prediction or a causal claim.

> **READ THE "WHERE IS THE RISK" SECTION CAREFULLY**
>
> It lists the worst market, region and category — but the page immediately warns that only the shipping-mode row is a real finding. The others sit within about a percentage point of the overall rate and none survives statistical testing. They are shown for completeness, not as targets for action.

E. WHAT DOES THE RESULT TELL US?

The four recommendations are: re-set the Second Class delivery promise; adopt risk-based shipping assignment; do not invest in destination- or product-level delay forecasting; and review service commitments before changing routing. Notice that one of the four is a recommendation to spend nothing — a useful finding in its own right.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It is the bridge from analysis to action. Each recommendation names what to do, what it is expected to achieve, and how confident you can be — so a decision-maker can weigh them without needing to interpret a single chart.

G. IMPORTANT THINGS TO REMEMBER

- Everything is generated from the computed results, so it always matches the other tabs.
- Recommendations are supported by evidence but do not guarantee outcomes.
- The economic figures move with the assumption sliders on Tab 9.

TAB 12 OF 12

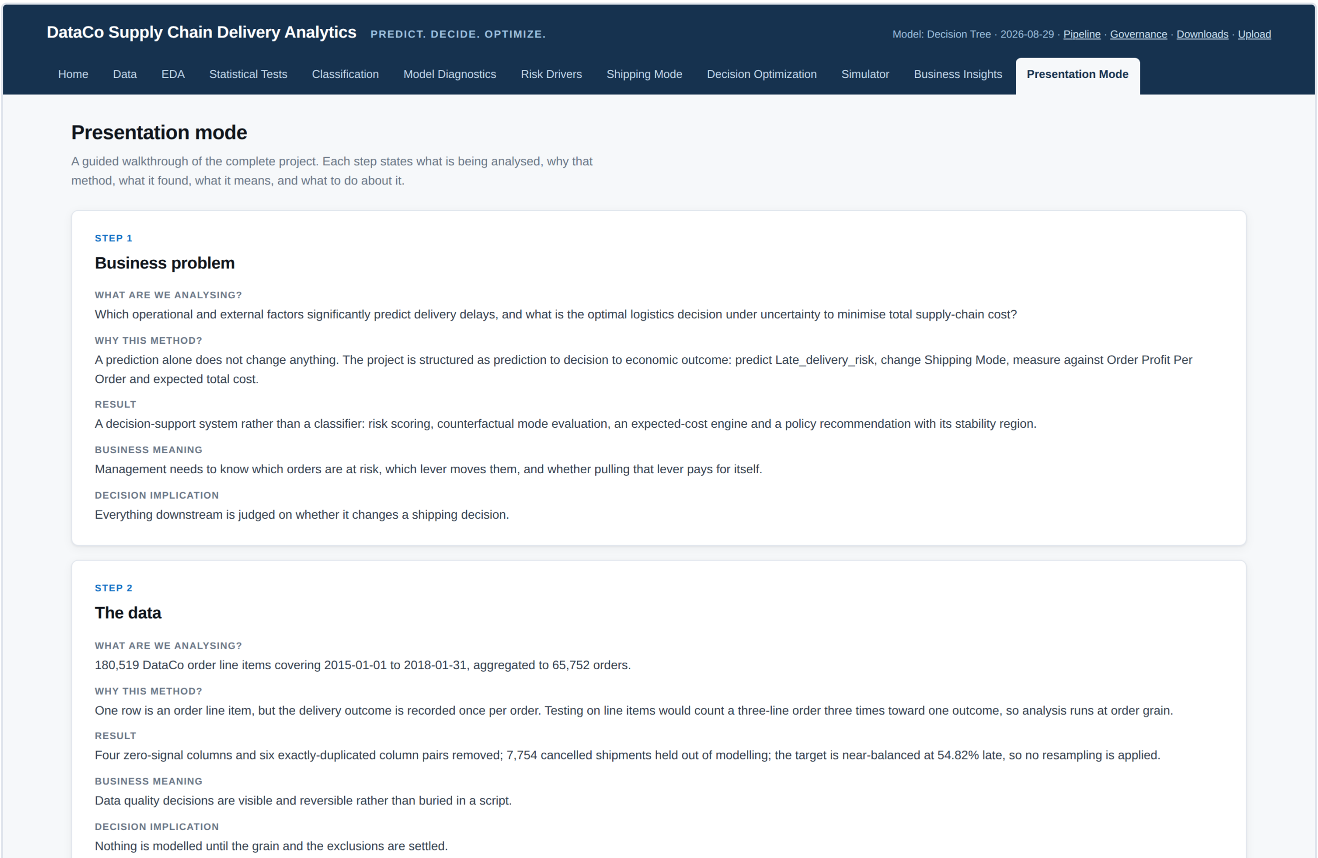
Presentation Mode

A. WHAT IS THIS TAB?

A guided walkthrough of the entire project in twelve numbered steps, designed to be presented from directly. Each step answers the same five questions in the same order, so the argument builds consistently from the business problem to the final recommendation.

B. WHY DO WE NEED THIS TAB?

So that someone can understand the whole project without reading the code or clicking through every tab. It is the page to have open when presenting.



> **Presentation Mode.** Each step answers the same five questions: what, why, result, business meaning, decision implication.

C. WHAT DO WE SEE HERE?

Twelve numbered panels, each with five labelled sections:

QUESTION	WHAT IT COVERS
What are we analysing?	The subject of this step
Why this method?	Why this approach rather than another
Result	What was actually found, with the real figures
Business meaning	What a manager should take from it
Decision implication	What action it points toward

The twelve steps run: business problem, the data, delivery performance, statistical significance, model comparison, validation and leakage, risk drivers, the shipping-mode trade-off, decision optimization, policy comparison and sensitivity, live demonstration, and management recommendations.

D. HOW SHOULD I READ IT?

Top to bottom, in order. The "Why this method?" line is the one most likely to be examined in a viva — it is where the methodology is justified.

E. WHAT DOES THE RESULT TELL US?

It contains the same findings as the rest of the application, arranged as an argument rather than as a set of tools.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It shows the reasoning, not just the conclusion. A recommendation is far easier to accept when you can see how it was reached and what was ruled out along the way.

G. IMPORTANT THINGS TO REMEMBER

- Step 11 points to the live demonstration. Have the Simulator and Decision Optimization tabs open in other browser tabs before presenting.
- The closing caveat about assumptions and causation is part of the argument, not a disclaimer to skip.

Pipeline

Reached from the "Pipeline" link in the top-right corner.

A. WHAT IS THIS PAGE?

A single diagram showing all sixteen stages the analysis passes through, from the raw data file to the final business decision, with a short description of what happened at each stage.

B. WHY DO WE NEED THIS PAGE?

It answers "what did you actually do?" in one screen. It also makes clear that the order of operations was deliberate — cleaning before exploring, screening before modelling, validating before deciding.

C. WHAT DO WE SEE HERE?

Sixteen numbered boxes connected by arrows: raw data, data quality, grain reconciliation, EDA, feature engineering, leakage screening, train/test split, model training, cross validation, model evaluation, statistical significance, interpretability, late-risk prediction, shipping-mode scenario analysis, expected-cost optimization, and business decision.

D. HOW SHOULD I READ IT?

Top to bottom. Each box names a stage and states what was actually done, including specific figures such as the memory reduction from 350 MB to 17 MB.

E. WHAT DOES THE RESULT TELL US?

That the analysis followed a disciplined sequence in which each stage depends on the one before it, rather than jumping straight to modelling.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It supports the credibility of the conclusions. It also gives a reviewer a map for asking targeted questions about any single stage.

G. IMPORTANT THINGS TO REMEMBER

- It describes the process, not the findings — use it to explain method, not results.
- Stages 6 and 7 (leakage screening and the grouped split) are the two that most affect how trustworthy the final numbers are.

Governance

A. WHAT IS THIS PAGE?

An honest account of what this analysis cannot do. It lists the limitations, explains what would need monitoring if the model were used in practice, and records the settings that make the results reproducible.

B. WHY DO WE NEED THIS PAGE?

Every model has limits, and stating them openly is a mark of quality rather than weakness. In an academic setting this is often the page that separates a careful project from an overconfident one.

C. WHAT DO WE SEE HERE?

- **Limitations** — two panels covering the data and inference constraints, and what this dataset can and cannot answer.
- **Model monitoring** — seven measures to track in production, each with its current baseline and what a change in it would mean.
- **Reproducibility** — the random seed, when the results were generated, the split strategy and the chosen model.

D. HOW SHOULD I READ IT?

The most important limitation is stated plainly: **the delay process in this dataset is close to being determined entirely by the shipping method.** With shipping mode removed, all 36 remaining factors together score 0.535 against a chance level of 0.500.

> **WHAT THIS MEANS FOR YOUR CONCLUSIONS**
>
> Conclusions about which factors drive delay describe this dataset's structure. They should not be presented as a general finding that geography or product type never affect delivery performance in the real world.
>
> What does transfer is the method: screening out information that would not be available at decision time, testing at the right level of aggregation, reporting effect sizes alongside p-values, and pricing the reliability-versus-speed trade-off explicitly instead of just minimising the late rate.

E. WHAT DOES THE RESULT TELL US?

That the findings are solid within this dataset, and that the boundaries of what can be claimed have been mapped rather than glossed over.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It prevents overreach. A manager reading this knows which recommendations rest on measurement and which rest on assumption, and what would need watching if the model were deployed.

G. IMPORTANT THINGS TO REMEMBER

- Reading this page before presenting means you will not be caught out by the obvious challenge — you will have raised it first.
- The privacy note is a real operational point: the source file contains customer names, addresses and coordinates even though no model uses them.

Downloads

A. WHAT IS THIS PAGE?

Nine buttons, each producing a spreadsheet file of results you can open in Excel.

B. WHY DO WE NEED THIS PAGE?

So results can leave the application — for an appendix, a slide, or someone who wants to check a number themselves.

C. WHAT DO WE SEE HERE?

EXPORT	CONTENTS
Model comparison	All eight scores for each of the four models
Feature importance	All three importance measures
Statistical significance	Every test with statistic, p-value, effect size and interpretation
Shipping-mode economics	Orders, late rate, promised and actual transit, profit per method
Decision / scenario analysis	Policy comparison with costs and mode mix
Break-even analysis	The upgrade ladder with break-even probabilities
EDA summary	Late rate, order count and profit for every level of every dimension
Predictor availability	Every column, whether it was used, and why
Diagnostic report	Logistic coefficients, VIF and calibration

D. HOW SHOULD I READ IT?

Each file matches exactly what is displayed on the corresponding tab — they are generated from the same stored results, so a downloaded figure can never disagree with the screen.

E. WHAT DOES THE RESULT TELL US?

Nothing new; it is a delivery mechanism for findings already presented.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It makes the analysis auditable. "Predictor availability" is the single most useful file to have to hand, because it answers the common challenge of whether anything was quietly left out.

G. IMPORTANT THINGS TO REMEMBER

- Downloads reflect the assumptions stored when the analysis was prepared, not any changes you have just made with the sliders.
- The files are plain CSV and open directly in Excel.

Upload

A. WHAT IS THIS PAGE?

A way to load a different data file and have the application check it before anything is run on it.

B. WHY DO WE NEED THIS PAGE?

So the application is not permanently tied to one file. If a newer extract arrives, this page checks it has the required structure before you rebuild the analysis.

C. WHAT DO WE SEE HERE?

A file selector and a "Upload and validate" button. After uploading, a status list appears confirming that the file loaded, the target was found, the decision variable was found, the economic variables were found, and the leakage and personal-data screening completed. Any problems are listed separately. The right-hand panel shows details of the currently loaded dataset.

D. HOW SHOULD I READ IT?

Ticks are checks that passed. Anything in the "Issues found" box needs attention before the data can be used — most often a missing required column.

E. WHAT DOES THE RESULT TELL US?

Whether a new file is structurally compatible. It does not analyse the file — the training step must be re-run separately, as the page states.

F. WHY IS THIS USEFUL FOR MANAGEMENT?

It shows the analysis is repeatable on fresh data rather than a one-off exercise.

G. IMPORTANT THINGS TO REMEMBER

- Uploading does not retrain anything. The training step must be re-run afterwards.
- The application checks structure, not accuracy — it cannot tell whether the values are correct.

4. Glossary of terms

Every term below appears somewhere in the application. Each is given in three parts: what it is, in plain words, and why it matters here.

TERM	SIMPLE MEANING	WHY IT MATTERS HERE
Late delivery risk	Whether an order arrived after the date promised. Yes or no.	The thing being predicted. 54.8% of orders were late.
Target variable	The thing a model is trying to predict.	Here it is late delivery risk.
Predictor / feature	A piece of information used to make the prediction.	37 are used, such as shipping method, destination and order value.
Classification	Sorting things into categories rather than predicting a number.	Sorting orders into "likely late" and "likely on time".
Regression	Predicting a number rather than a category.	Used to estimate the profit of an order.
Grain	What one row of data represents.	Rows are order lines but deliveries are per order, so analysis runs at order level.
Training / test data	Data used to build the model, versus data held back to check it.	138,493 rows train it; 34,272 unseen rows test it.
Data leakage	Accidentally using information that would not be available when the decision is made.	Three columns reveal the answer and are excluded. Including them fakes 99% accuracy.
Probability	A number from 0 to 1 expressing how likely something is.	The model outputs one per order; the decision engine multiplies it by cost.
Accuracy	Share of predictions that were correct.	0.7172. Misleading alone — it hides which kind of error was made.
Precision	When the model says "late", how often it is right.	0.8821. High precision means few false alarms.
Recall (sensitivity)	Of all genuinely late orders, how many were caught.	0.5870. Low recall means late orders slip through unflagged.
Specificity	Of all on-time orders, how many were correctly left alone.	The mirror image of recall.
F1 score	A single number balancing precision and recall.	0.7049. Useful for comparing models at a glance.
Confusion matrix	A count of the four possible outcomes: correct and incorrect, each way.	Separates false alarms (1,547) from missed late orders (8,145).
False positive	Predicted late but arrived on time — a false alarm.	Costs an unnecessary upgrade. Bounded and known in advance.
False negative	Predicted on time but arrived late — a missed case.	The expensive error: no warning, no action, surprised customer.

ROC-AUC	How well the model separates risky from safe orders across all cut-offs. 0.5 is guessing.	0.7680, against 0.535 without shipping mode.
PR-AUC	Similar, but focused on how well it finds the late orders specifically.	0.8260. More informative when one outcome matters more.
Threshold	The probability cut-off at which an order is called "risky".	Adjustable on the slider. Lower catches more late orders but raises false alarms.
Youden's J	A statistic suggesting a balanced cut-off.	0.4826 at a cut-off of 0.71 — a diagnostic, not the business-optimal choice.
Calibration	Whether "70% likely" really does happen about 70% of the time.	Essential, because the decision engine turns probabilities into money.
Brier score	A single measure of how accurate the probabilities are. Lower is better.	0.1776.
P-value	The chance a difference this large could be a fluke. Below 0.05 is usually called significant.	Says whether a difference is real, not whether it is big enough to matter.
Statistical significance	The difference is probably not down to chance.	Only one of 18 factors reaches it: shipping mode.
Effect size	How big a difference is, in a way that does not grow with sample size.	The measure that actually matters commercially. Shipping mode scores 0.481; the rest are near zero.
Cramér's V	The particular effect size used for category comparisons. 0 means no association.	Lets factors be ranked by importance rather than by p-value.
Correlation	Whether two measures move together. From −1 to +1.	Lateness and profit correlate at −0.0056: essentially unrelated.
Feature importance	How much the model relies on each piece of information.	Shipping mode 0.2665; the next factor 0.0493; the rest near zero.
Permutation importance	Scramble one column and see how much worse the model gets.	Measured on unseen data, so it cannot reward memorisation.
Odds ratio	How much a factor multiplies the odds of an outcome. Above 1 raises them; below 1 lowers them.	Makes the statistical model readable in business terms.
VIF	Checks whether two predictors are telling the model the same thing.	Flags that the promised window and the shipping method are effectively one variable.
Expected cost	The average cost of a decision if you made it many times.	The basis of the whole decision engine.

Break-even point	The point at which two options cost the same.	Tells you how likely a delay must be before upgrading pays for itself.
Sensitivity analysis	Changing an assumption to see whether the answer changes.	Shows how robust the recommendation is to the numbers you cannot measure.
Scenario assumption	A number supplied by the analyst because the data does not contain it.	All cost figures. The dataset has no shipping-cost field at all.
Association	Two things tend to occur together.	What this analysis can demonstrate.
Causation	One thing actually makes the other happen.	What this analysis cannot demonstrate, because shipping methods were never randomly assigned.

5. How all the tabs fit together

Each tab answers one question, and the questions build on each other. Read down the table and you have the whole argument.

APPLICATION SECTION	MAIN QUESTION ANSWERED	BUSINESS PURPOSE
Home	What is happening?	Understand the overall situation
Data	What information do we have?	Understand and trust the evidence
EDA	What patterns exist?	Discover where the delays are
Statistical Tests	Are those differences meaningful?	Validate the evidence, avoid chasing noise
Classification	Can we identify risky orders in advance?	Predict future risk honestly
Model Diagnostics	How good and how trustworthy is the model?	Know what kind of mistakes it makes
Risk Drivers	Why are some orders riskier?	Understand which factors matter
Shipping Mode	What can we actually change?	Identify the decision lever
Decision Optimization	What should we choose, and what is it worth?	Evaluate the trade-off and support the decision
Simulator	What happens for this particular order?	Explore scenarios order by order
Business Insights	What should management do?	Turn analysis into action
Presentation Mode	How does the whole argument hang together?	Explain the project end to end
Pipeline	What steps were followed?	Demonstrate a disciplined method
Governance	What are the limits of this analysis?	Set honest expectations
Downloads	How do I get the results out?	Make the findings auditable
Upload	Can this be repeated on new data?	Show the analysis is repeatable

The story in one paragraph

More than half of all deliveries arrive later than promised. Looking for the cause, we find that geography, product type, customer type and order size explain essentially none of it — the differences between markets are around one percentage point and fail every statistical test. What does explain it is the shipping method, which spans a 57-point range. Digging further, the reason is not that some methods are slower: Second Class and Standard Class deliver in the same time, but Second Class promises half as long, so it breaks its promise twice as often. That points to two levers — re-setting delivery promises, which costs nothing operationally, and assigning shipping methods order by order based on predicted risk and order value, which under our cost assumptions is about 12% cheaper than a single fixed policy while roughly halving the expected late rate.

6. How to explain the application in 5 minutes

A spoken script for a classroom presentation. Each step names the tab to have on screen. The words are written to be said aloud, not read.

Step 1 — The business problem (tab: Home)

"Our question is: which factors predict delivery delays, and what shipping decision minimises total supply chain cost under uncertainty?

More than half of this company's orders — 54.8% — arrive later than promised. But this project is not just a prediction exercise. Predicting a late delivery is only useful if we can do something about it, and if we know whether that something is worth the money. So we built three connected pieces: we **predict** late delivery risk, we **decide** the shipping mode, and we judge it on **profit and expected cost**."

Step 2 — The data (tab: Data)

"We have three years of DataCo orders — 180,519 order lines, covering 65,752 actual deliveries. That distinction matters: a customer buying two products creates two rows but one delivery. So we analyse at order level, otherwise we would count a single late delivery several times.

We removed four columns that were entirely empty or identical throughout, six that were exact duplicates of other columns, and all personal data. There are no duplicate rows and very few missing values."

Step 3 — The evidence (tabs: EDA, then Statistical Tests)

"Here is the late rate by market. Europe 55.0%, Pacific Asia 55.3%, LATAM 54.4%, USCA 54.8%, Africa 54.1%. They are all the same. Same story for region, product category and customer segment.

Now here is late rate by shipping mode: 38% to 95%. That is a 57-point spread.

We tested this formally — 18 factors, with corrections for testing many things at once. **Exactly one is significant: shipping mode**, with a large effect size of 0.48. Every other factor fails. And that is a strong result, because with 63,000 orders these tests would detect even a small real difference. Finding nothing means there is essentially nothing to find."

Step 4 — The prediction (tabs: Classification, Model Diagnostics)

"We trained four models and tested them on 34,272 orders they had never seen. The best scores 0.768, where 0.5 is a coin toss.

That number is lower than most published analyses of this dataset, which report around 99%. The difference is deliberate. Three columns in this data reveal the answer — delivery status, actual shipping days, and the shipping date. If you leave them in, the model is not predicting, it is reading the answer. We excluded all three.

Here is our check: with shipping mode removed, every remaining factor together scores 0.535. Barely above chance. That confirms both that the screening worked, and that shipping mode is carrying essentially all the signal."

Step 5 — The decision (tab: Shipping Mode)

"So shipping mode is the driver, and conveniently it is also the one thing management controls. But look at why it matters.

Standard Class promises four days and takes 3.99. Second Class promises two days and takes 4.00. **Identical transit time.** Yet Standard Class is late 40% of the time and Second Class 80%.

The entire gap is the promise, not the operation. First Class is the extreme case: it always takes exactly two days but promises one, so it is late 100% of the time. It is not slow, it is over-promised.

That gives us a lever that costs nothing: re-set the Second Class promise to match Standard Class, and its late rate should fall by around 40 points across 12,256 orders, without changing a single carrier or route."

Step 6 — The economics (tab: Decision Optimization)

"Now the cost. And I want to be upfront: **this dataset contains no shipping-cost data at all.** So rather than inventing freight costs and presenting them as findings, we built a transparent model where every cost is an assumption you can change — and we show how far you would have to move them before the answer changes.

Expected cost for each option is freight, plus the chance of being late multiplied by what lateness costs, plus the cost of capital in transit, plus the value of the days saved.

One honest finding here: if you use only freight plus lateness cost — the obvious formula — Standard Class wins for 99.8% of orders under any assumptions, because it is both the cheapest and the most reliable. The optimisation answers itself. The reason is that a simple late/on-time measure cannot see that Same Day delivers in half a day against Standard's four. Adding the time terms restores a genuine trade-off."

Step 7 — The recommendation (tab: Business Insights)

"Three recommendations.

First, re-set the Second Class delivery promise. Same operation, achievable promise, around 40 points off its late rate at no operational cost. This is the strongest finding because it rests on measured transit times, not assumptions.

Second, assign shipping mode order by order based on predicted risk and order value. Under our cost assumptions that is about 12% cheaper per order than a single fixed policy, and roughly halves the expected late rate.

Third, do not invest in destination- or product-level delay forecasting. We tested for it three separate ways and the signal is not there. Knowing where not to spend is worth as much as knowing where to spend.

Two caveats. The cost figures are assumptions, clearly labelled, and we show the sensitivity. And because shipping methods were never randomly assigned, we can say shipping mode is strongly associated with late delivery — we do not claim it causes it. That is why the recommendation about re-setting promises is the strongest one: it rests on measured transit times, not on a modelled counterfactual."

IF YOU HAVE ANOTHER TWO MINUTES: THE LIVE DEMO

Open the **Simulator**. Change the shipping mode and watch the risk meter move sharply. Then change the destination market and watch it barely move — the headline finding, demonstrated live.

Then open **Decision Optimization**, set both time sliders to zero, and watch the recommendation collapse onto one shipping method. Raise the value of speed and watch it flip back. That is uncertainty in the business question, made visible.

Questions you should expect

QUESTION	ANSWER
"Only one significant variable? Did the analysis fail?"	No — three independent methods agree: the hypothesis tests, the model ablation, and permutation importance. Finding one strong driver and ruling out sixteen others is a result.
"Why is your accuracy lower than published results?"	Because we removed the three columns that reveal the answer. Those results are scoring the outcome against itself.
"Why only 18 tests when the model has 37 features?"	Several features are alternative encodings of one factor, and some have too many categories for a valid test. The ablation covers all 36 at once and finds 0.535.
"Where did the shipping costs come from?"	They are stated assumptions — the dataset has none. That is why the sensitivity analysis exists.
"Does shipping mode cause delays?"	We cannot claim that. Assignment was not randomised, so we report association and model-implied scenarios.